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Standard Test Methods for Rating Portal Stability of Enchanted Moon-Glass¹

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¹ Prepared solely as synthetic material for information-extraction training. All entities, apparatus, units, observations, and precision statements are invented.

1. Scope

1.1 These test methods cover procedures for assessing the resistance of enchanted moon-glass vessels to loss of stored whisper content and change in apparent hue when the vessels are passed through a controlled paired portal.

1.2 Test Method A is primarily intended to rate the stability of a single ordered whisper imprint in vials having an enchantment density from 40 to 80 aether marks per cubic handspan (AM/ch3), while Test Method B is primarily intended to rate the stability of a six-cell whisper lattice in similarly charged vessels. Test Method B is not considered suitable for vessels containing a resident or self-editing enchantment.

1.3 These test methods are used to evaluate whether portal stability is adequate for the user's application. They do not distinguish between high levels of stability for which more sensitive methods of measurement are required.

1.4 In multivoice imprints, whisper loss may occur between adjacent rune layers so that the stability of the imprint at the moon-glass interface is not determined.

1.5 The values stated in millilitres, seconds, degrees Celsius, and the fictional units AM/ch3, motes/m3, and spectral degrees are to be regarded as the units of this document. The fictional units have no physical counterpart.

1.6 *This document does not purport to address real safety concerns. Within the fictional scenario, it is the responsibility of the user to establish appropriate portal containment, echo-control, and shutdown practices and to determine the applicability of limitations prior to use.*

2. Referenced Documents

2.1 Fictional Guild Documents:

GP-12 Practice for Conditioning Enchanted Containers in Quiet Rooms

AT-3 Terminology Relating to Echo-Bearing Materials

PSC-7 Practice for Containment of Low-Energy Transit Rings

RM-9 Test Method for Assigning Spectral Degrees to Luminous Artifacts

WT-4 Practice for Producing Whisper Imprints of Uniform Density

NOTE 1-The referenced documents are fictional and are included only to reproduce the citation relationships ordinarily found in a technical test method.

3. Terminology

3.1 Definitions of Terms Specific to This Document:

3.1.1 *aether mark, n* -a fictional unit representing one stable increment of stored enchantment.

3.1.2 *batch, n* -as it pertains to moon-glass, a unique production run cast during one lunar phase.

3.1.3 *blemish, n* -an obvious surface flaw, such as silvering, a hairline crack, or a persistent luminous spot.

3.1.4 *chromatic drift, n* -the smaller angular difference between initial and final hue signatures, expressed in spectral degrees.

3.1.5 *environmental conditions, n* -the characteristics of the immediate surroundings during the tests, including temperature, relative humidity, and background mana.

3.1.6 *lattice whisper, n* -six ordered rune groups arranged as a two-by-three imprint and stored as one test pattern.

3.1.7 *portal cycle, n* -one complete outbound and return passage through paired transit rings.

3.1.8 *quiet room, n* -an enclosure maintained at or below 2.0 background motes/m3 and isolated from speech, bells, music, and divination.

3.1.9 *spectral degree, n* -one of 360 equal positions on the fictional circular hue lattice.

3.1.10 *whisper imprint, n* -an ordered sequence of neutral runes stored in a moon-glass test specimen.

3.1.11 *test specimen, n* -the moon-glass vessel whose portal stability is of interest.

4. Summary of Test Methods

4.1 *Test Method A* -A single 12-rune whisper is placed in a conditioned vial, the initial hue is recorded, and the vial is exposed to five portal cycles. The recovered sequence and final hue are compared with the initial values, and portal stability is assessed qualitatively on a 0 to 5 scale.

4.2 *Test Method B* -A six-cell whisper lattice is placed in a conditioned vial and exposed to five portal cycles. Each lattice cell is examined for loss, substitution, or migration of runes, and portal stability is assessed qualitatively on a 0 to 5 scale.

5. Significance and Use

5.1 In order for a moon-glass vessel to fulfill its function of retaining a message during transit, the stored whisper must remain associated with the glass and in its assigned order. Because glass charge, vessel geometry, imprint density, and portal alignment can have a pronounced effect on retention, a method for comparing different vessels or portal settings is of considerable usefulness in the fictional trade.

5.2 These test methods are limited to evaluating lower levels of portal stability. The rating range from 0 to 5 reflects the inability of these test methods to make fine distinctions between high levels of whisper retention. Users shall not use intermediate values for ranking results within either method.

5.3 Extremes in temperature, relative humidity, or background mana may affect the apparent enchantment density of the specimen and the response of the whisper comparator.

5.4 A given portal setting may not affect all moon-glass compositions equally. As such, no single setting is likely to be suitable for comparing every vessel composition, charge density, and closure design.

5.5 Operators performing these test methods shall be trained in portal timing and in interpretation of whisper-comparator results. The precision of a result depends substantially upon the operator's ability to maintain portal dwell time, recovery interval, inquiry-rune presentation, and specimen orientation in a consistent manner.

NOTE 2-These methods provide ordinal ratings. They do not determine the force, energy, or mechanism responsible for whisper loss.

TEST METHOD A-SINGLE-WHISPER TRANSIT TEST

6. Apparatus and Materials

6.1 *Paired Transit Rings* -Two interlocked low-energy rings having a clear aperture of 300 +/- 5 mm, an adjustable fringe setting, and an emergency closure time not exceeding 0.8 s.

6.2 *Specimen Cradle* -A nonreflective holder that supports the vial upright without contacting more than 15 % of its external surface.

6.3 *Whisper Comparator* -An instrument capable of presenting one inquiry rune and decoding the 12 positions of the returned whisper.

6.4 *Circular Hue Lattice* -A comparator having 360 spectral-degree divisions and a repeatability of 0.2 spectral degree or better.

6.5 *Aether Densitometer* -A device readable to 0.5 AM/ch3 over the range from 0 to 100 AM/ch3.

6.6 *Reference Echo Crystal* -A stable fictional comparison artifact encoded with the 12-rune reference sequence LUM-ARA-VEK-SIL-ORA-NEM-TAL-IRI-MOR-PEL-UNE-DRA.

7. Test Specimens

7.1 When this method is used for a field simulation, the specimen is the vessel on which portal stability is to be evaluated.

7.2 For laboratory use select not fewer than three vessels from each moon-glass batch. The vessels shall have the same nominal volume, wall profile, closure condition, and charging history.

NOTE 3-Vessels having visible cracks, an illegible batch mark, an attached familiar, or a pre-existing whisper shall not be tested.

7.3 Condition specimens for 16 to 24 h at 21 +/- 2 deg C, 45 +/- 10 % relative humidity, and not more than 2.0 background motes/m3.

8. Procedure

8.1 Select an area of each vessel free of blemishes and minor surface imperfections. The specimen shall be clean, dry, and free from active charms.

8.2 Place the specimen in the cradle with its identification mark facing the operator. Measure initial enchantment density at three equally spaced positions and record the mean.

8.3 Apply the 12-rune reference whisper once over 12.0 +/- 0.2 s. After 60 +/- 5 s, verify an enchantment density of 60 +/- 3 AM/ch3.

8.4 Measure the initial hue signature at three positions 120 deg apart and record the circular mean, H0, to the nearest 0.1 spectral degree.

8.5 Present the inquiry rune and establish that all 12 reference runes are returned in the correct order. If the reference whisper is not returned, reject the specimen and select another vessel.

8.6 Set the transit rings to fringe setting 4 and a transit duration of 2.0 +/- 0.1 s in each direction. Pass the specimen outward and, after 30 +/- 2 s, return it through the paired ring. Repeat until five portal cycles have been completed.

8.7 After the fifth cycle, allow the specimen to recover for 10.0 +/- 0.2 min. Measure the final hue signature, H1, and present the inquiry rune once.

8.8 Inspect the returned sequence for omission, substitution, displacement, repetition, or unsolicited runes. Rate the result in accordance with the scale illustrated in Fig. 1.

CLASSIFICATION OF PORTAL STABILITY RESULTS FOR TEST METHOD A

Classification	Description of observed result
5A	All 12 runes are recovered in order; chromatic drift is not greater than 1.0 spectral degree.
4A	One outer rune is faint or delayed; no rune is omitted; chromatic drift is not greater than 3.0 spectral degrees.

Classification	Description of observed result
3A	One rune is omitted or substituted, or two runes are displaced; chromatic drift is not greater than 6.0 spectral degrees.
2A	Two or three runes are omitted, substituted, or displaced; no unsolicited sequence is emitted.
1A	Four to eight runes are not recovered in their assigned positions, or one unsolicited sequence is emitted.
0A	Fewer than four reference runes are recovered, or the vessel emits a persistent unrelated whisper.

FIG. 1 Classification Of Portal Stability Results For Test Method A

8.9 Repeat the test on two other specimens from the same batch. For large batches, make sufficient tests to ensure that the evaluation is representative of the batch.

8.10 Examine the portal alignment after each series. Discard results obtained during a portal interruption, an out-of-tolerance recovery interval, a comparator fault, or visible specimen fracture.

9. Report

9.1 Report the moon-glass batch, vessel volume, wall profile, closure condition, and conditioning history.

9.2 Report the number of tests, the individual classifications, their median, and their range.

9.3 Report H0, H1, chromatic drift, initial and final enchantment density, and the recovered rune at each of the 12 positions.

9.4 Report the portal setting, number of completed cycles, recovery interval, apparatus identifiers, and any visible silvering, cap rotation, unsolicited rune, or luminous trail.

9.5 For field simulations, report the location and the environmental conditions at the time of testing.

10. Precision and Bias

10.1 In a fictional intertower study of Test Method A, operators in six towers each made one determination on three specimens from each of three invented moon-glass batches covering a wide range of portal stability. Based on the synthetic results, the following criteria should be used for judging the acceptability of ordinal ratings at the 95 % confidence level:

10.1.1 *Repeatability* -Provided portal stability is uniform within a batch, two results obtained by the same operator should be considered suspect if they differ by more than one rating unit.

10.1.2 *Reproducibility* -Two results, each the median of triplicate tests and obtained by different operators, should be considered suspect if they differ by more than 1.5 rating units.

10.2 Bias cannot be established because no accepted reference moon-glass vessel exists with an assigned true portal-stability rating.

TEST METHOD B-LATTICE-WHISPER TRANSIT TEST

11. Apparatus and Materials

11.1 The apparatus and materials shall be as described in 6.1 through 6.5.

11.2 *Lattice Imprinter* -A fictional imprinting device capable of placing six equal whisper cells in a two-by-three lattice without overlap.

11.3 *Lattice Guide* -A transparent nonreflective overlay divided into six equal cells for locating rune loss and migration.

11.4 *Magnifying Viewer* -An illuminated magnifier capable of resolving faint or doubled runes within each lattice cell.

12. Test Specimens

12.1 Test specimens shall be as described in Section 7. The test area shall be sufficiently uniform that all six lattice cells receive the same nominal enchantment density.

13. Procedure

13.1 Condition and identify each specimen as described in 7.2 and 7.3. Establish quiet-room conditions before imprinting the lattice.

13.2 Place the specimen on a firm base and apply six whisper cells in two rows of three. Each cell shall contain the ordered runes LUM-ARA-VEK and shall be separated from adjacent cells by 5 +/- 1 mm.

13.3 Measure initial enchantment density at the center of each cell. No cell shall differ from the six-cell mean by more than 2 AM/ch3.

13.4 Record the initial hue signature and verify all three runes in each of the six cells. Reject a lattice having any missing, substituted, or displaced rune before exposure.

13.5 Expose the specimen to five portal cycles using the conditions specified in 8.6, and allow the specimen to recover as specified in 8.7.

13.6 Using the lattice guide and magnifying viewer, inspect each cell for rune loss, substitution, migration, or duplication. Rate the result in accordance with the scale illustrated in Fig. 2.

CLASSIFICATION OF PORTAL STABILITY RESULTS FOR TEST METHOD B

Classification	Description of observed result
5B	All six cells retain all three runes in the assigned order; no rune crosses a cell boundary.
4B	One faint rune is present; less than 5 % of the total lattice content is affected.
3B	One or two runes are lost or displaced; 5 % to 15 % of the lattice content is affected.
2B	Three to six runes are lost, displaced, or substituted; 15 % to 35 % of the lattice content is affected.
1B	Seven to twelve runes are affected, or an entire cell is lost; 35 % to 65 % of the lattice content is affected.
0B	More than 65 % of the lattice content is affected, or the cell boundaries are no longer distinguishable.

FIG. 2 Classification Of Portal Stability Results For Test Method B

13.7 Repeat the test on two other specimens from the same batch.

14. Report

14.1 Report the information required by 9.1, 9.4, and 9.5.

14.2 Report the number of tests, the individual classifications, their median, and their range.

14.3 Report the initial and final rune content of each lattice cell and identify each lost, substituted, migrated, or duplicated rune.

15. Precision and Bias

15.1 On the basis of a second fictional intertower study in which six towers made three determinations on two specimens from each of four invented batches, the following criteria should be used for judging ordinal ratings at the 95 % confidence level:

15.1.1 *Repeatability* -Results obtained by the same operator should be considered suspect if they differ by more than one rating unit for two determinations.

15.1.2 *Reproducibility* -Two results, each the median of duplicate or triplicate determinations and obtained by different operators, should be considered suspect if they differ by more than two rating units.

15.2 Bias cannot be established for this test method.

15.3 The apparatus, specimens, study design, and numerical limits described in this section are synthetic and shall not be interpreted as experimental findings.

16. Keywords

16.1 chromatic drift; enchanted container; lattice whisper; moon-glass; portal cycle; portal stability; single-whisper transit test; spectral degree; whisper retention

APPENDIX

(Nonmandatory Information)

X1. COMMENTARY

X1.1. Introduction

X1.1.1 Can portal stability be measured? Within the fictional premise, the answer is both yes and no. No method described here determines the actual physical strength of an enchantment-to-glass bond. The methods do provide an indication of relative stability under prescribed portal conditions.

X1.1.2 Practical portal tests are generally either implied or direct. An implied test observes changes after an exposure that resembles service. A direct test would attempt to measure a defined force or energy of rune separation. The present methods are implied tests and their results are expressed by discrete classifications.

X1.2. Selection of Test Method

X1.2.1 Test Method A is readily applied when preservation of one ordered message is of primary interest. Test Method B provides additional information about the location and extent of loss but requires more careful imprinting and interpretation.

X1.2.2 The two methods should not be treated as interchangeable. A 4A classification and a 4B classification describe different observations and do not represent equal quantities of retained whisper content.

X1.3. The Portal Test

X1.3.1 The portal test is affected by the glass, the stored whisper, the ring alignment, and the measurement system. Variation in any one of these elements can produce a change in classification even when the remaining elements are unchanged.

X1.3.2 Major limitations are the low sensitivity of the ordinal scale, the dependence upon operator interpretation, and the inability to identify whether a missing rune failed at the glass interface or within a layered whisper.

X1.3.3 Repeatability within one rating unit is expected in the fictional data model. The classification scales should therefore be used to identify substantial differences rather than to rank nearly equivalent specimens.

X1.4. Procedural Problems

X1.4.1 The final evaluation is visual and symbolic in nature. Different operators can disagree when a rune is faint, doubled, partially displaced, or delayed. Reference examples and consistent illumination reduce, but do not eliminate, this source of variation.

X1.4.2 Field simulations can lead to variation because temperature, humidity, background mana, portal alignment, and recovery time are more difficult to control than in a quiet-room test.

X1.4.3 When these test methods are used within their scope and are performed carefully, some insight into the approximate relative stability of fictional moon-glass vessels can be gained.

REFERENCES

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